

ISYE 6420 Homework 5

Question 1

Model:

$$BPD = \beta_0 + \beta_1 \sqrt{MA} + \epsilon$$

where $\epsilon \sim N(0, \sigma^2)$

β_0, β_1 priors were chosen to allow for flexibility with both having $\sigma = 10$ and mean being 0 and 1 respectively. σ was chosen to be an exponential distribution ($\lambda = 2$ for flexibility) to maintain positivity and with the assumption that the value is likely closer to zero.

Estimators with associated uncertainty

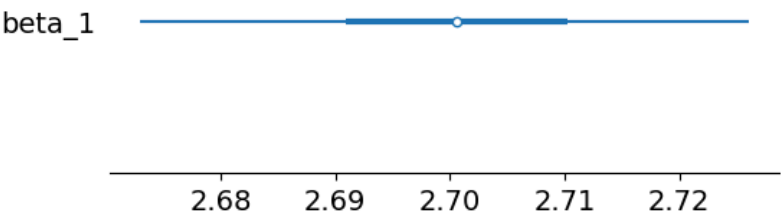
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd
beta_0	-7.456	0.071	-7.594	-7.328	0.002	0.001
beta_1	2.7	0.014	2.676	2.727	0	0
sigma	0.243	0.008	0.229	0.257	0	0

The variability response from the model can be captured from the R^2 value:

$$R^2 \approx 0.974$$

Hence, the model captures approximately 97% of the variability in BPD.

94.0% HDI

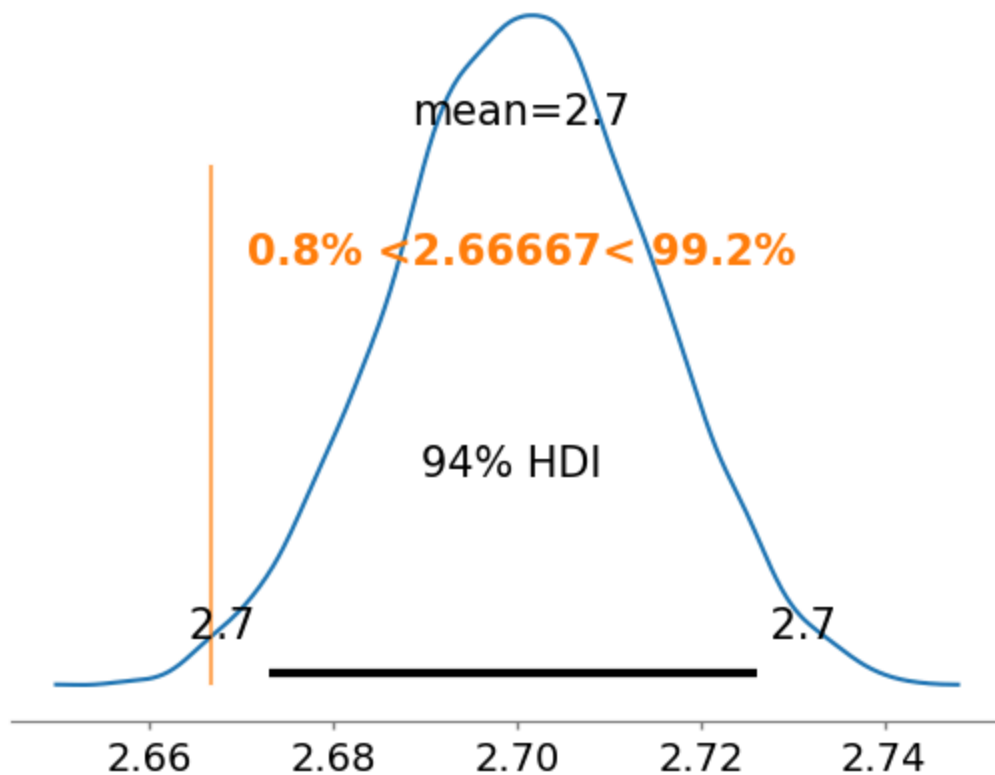


Hypothesis Testing at 8/3

Using the switch functionality in PyMC, a H_0, H_1 can be created.

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd
h1	0.992	0.09	1	1	0.002	0.011
h0	0.008	0.09	0	0	0.002	0.011

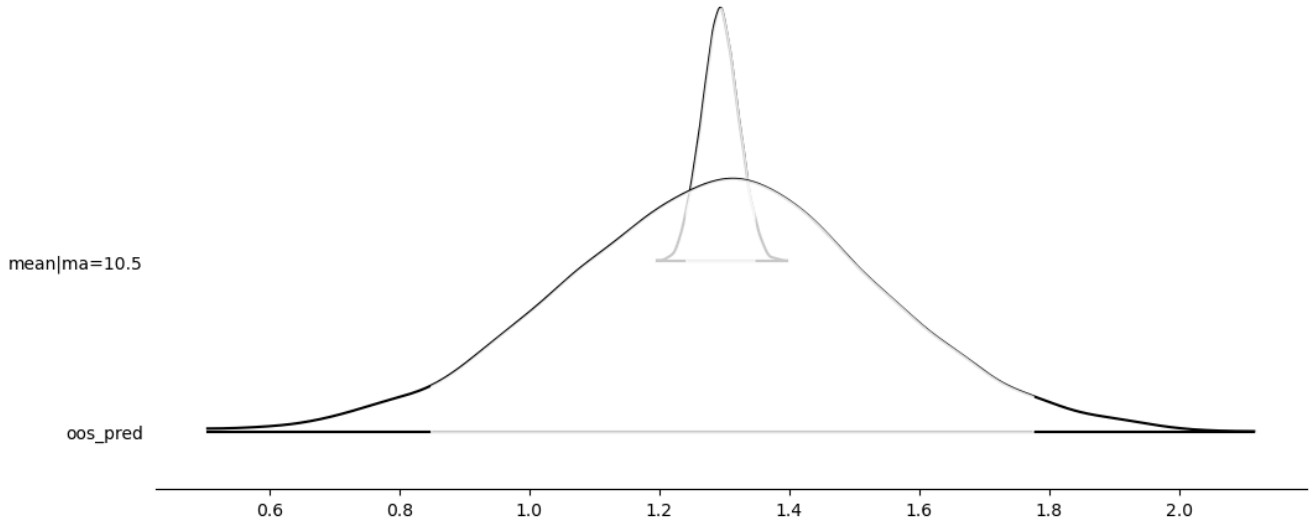
Posterior distribution of beta_1 with reference value 8/3



The posterior probability of $H_1 > 8/3$ being ≈ 0.99 shows a very strong indication that $\beta_1 > 8/3$ and the null hypothesis can be rejected.

Prediction at MA=10.5

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd
oos_pred	1.292	0.249	0.829	1.767	0.003	0.003
mean 10.5	1.294	0.028	1.242	1.346	0.001	0



As expected the interval for the individual response (the posterior predictive) has wider intervals than the mean response, as it also considers random, observational noise in the model.

Question 2

The priors on theta (1 to 4) were decided on the following assumptions:

θ_2 is limited between 0 and 1 so a uniform distribution with associated limits was decided as the simplest option.

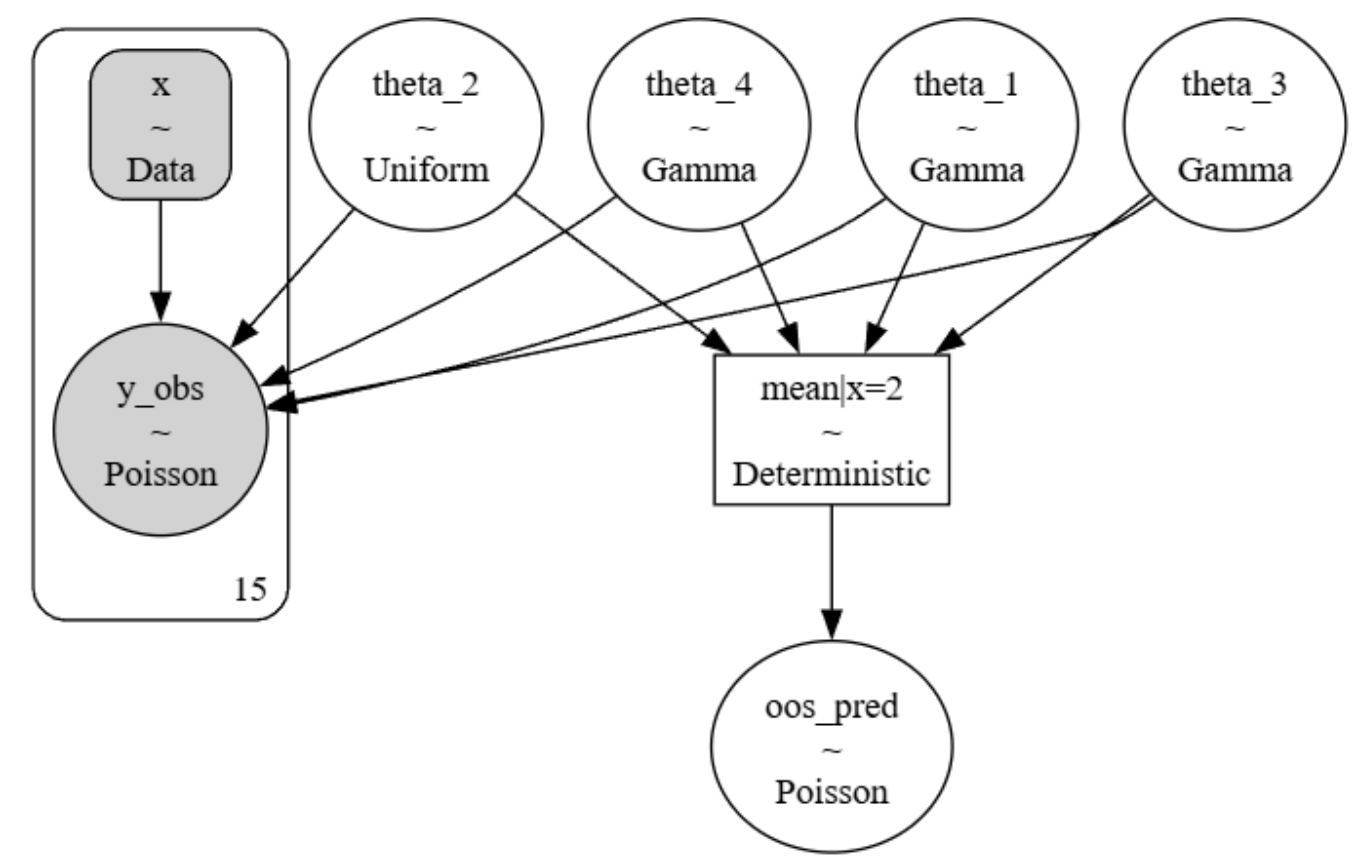
$\theta_1, \theta_3, \theta_4$ are positive values so a Gamma distribution was chosen for versatility and to satisfy the positivity requirement. The values for α, β were adjusted over multiple runs to minimise MCMC standard deviation, settling on the values as seen in the code file. θ_1 is an exceptional case as the resulting mean was ~ 100 , however the priors were not completely adjusted to accommodate for this large scale ($\alpha = 10, \beta = 0.5$), as the priors here are set to be weakly informative.

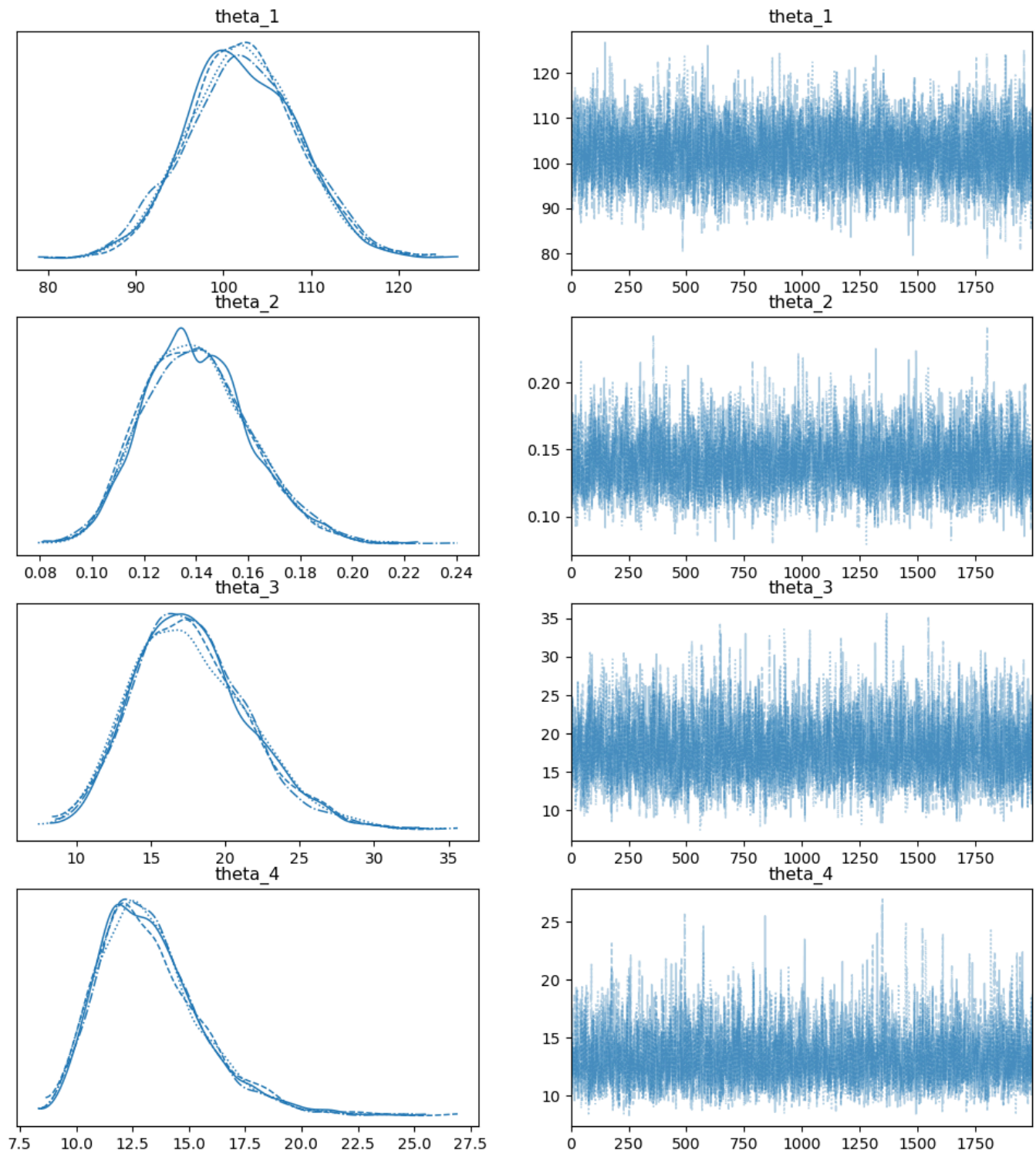
Regarding diagnostics, these values are provided below in the form of ESS and $\hat{r}$ values, which are >2000 and ≈ 1 respectively suggesting a stable model:

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta_1	0.093	0.071	4646	5007	1
theta_2	0	0	3281	5081	1
theta_3	0.064	0.045	3709	4434	1
theta_4	0.04	0.04	3640	3945	1

With the results of the model being:

	mean	sd	hdi_2.5%	hdi_97.5%
theta_1	102.319	6.353	90.082	114.597
theta_2	0.141	0.021	0.104	0.182
theta_3	17.732	3.936	10.46	25.701
theta_4	13.287	2.316	9.323	17.935

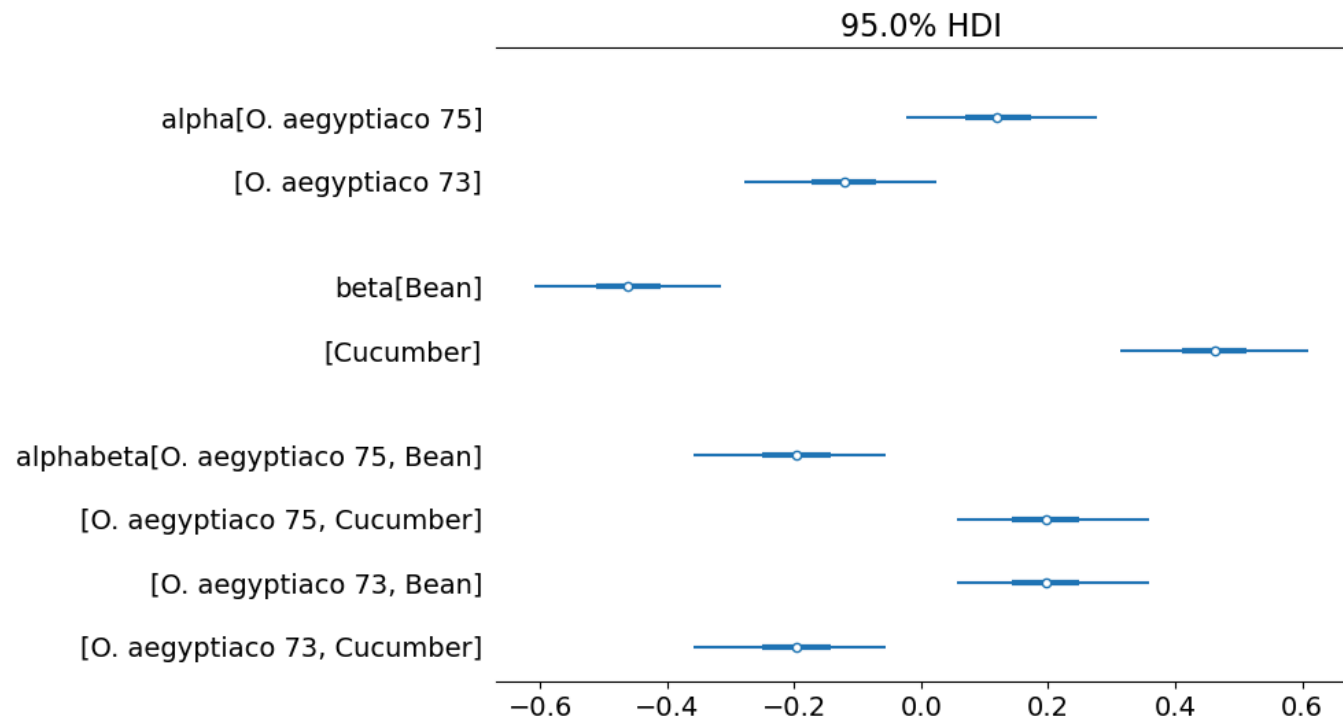
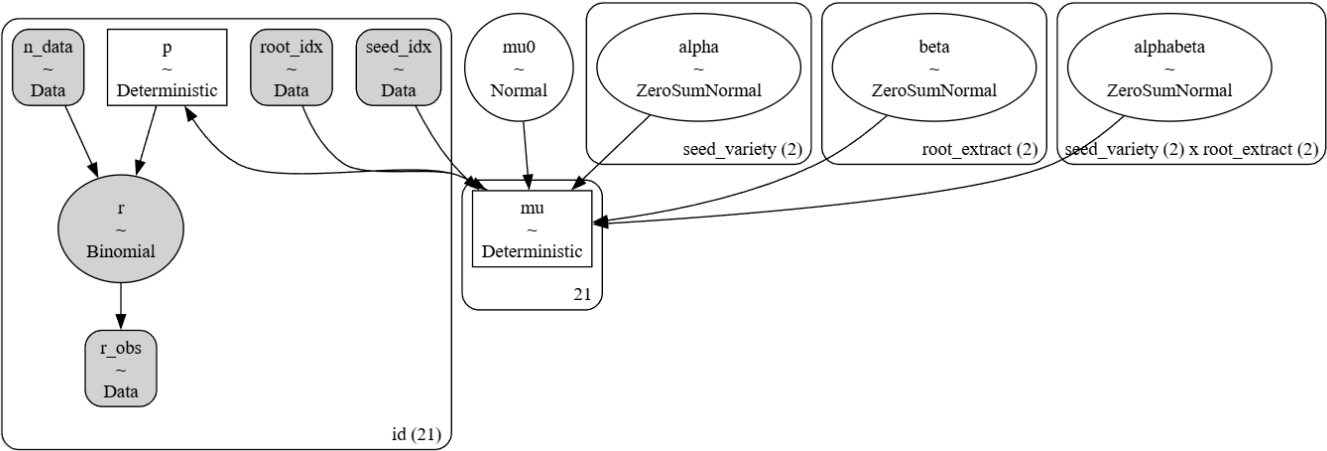


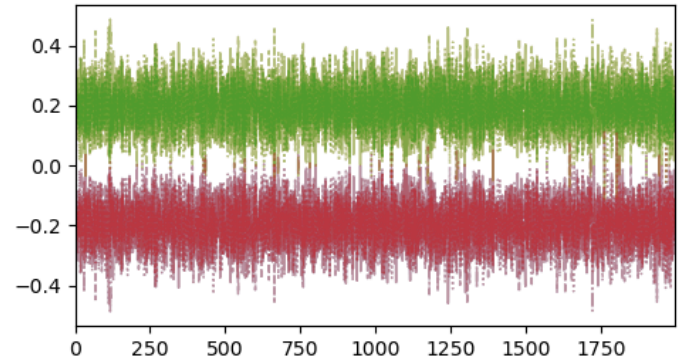
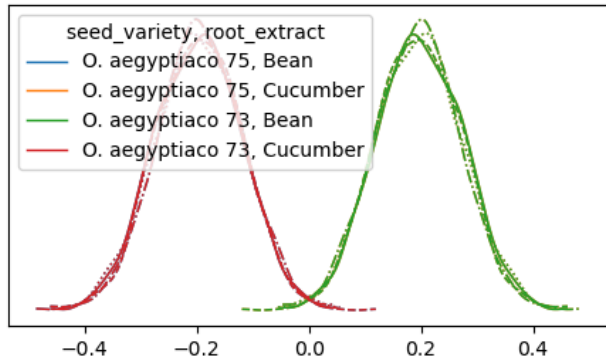
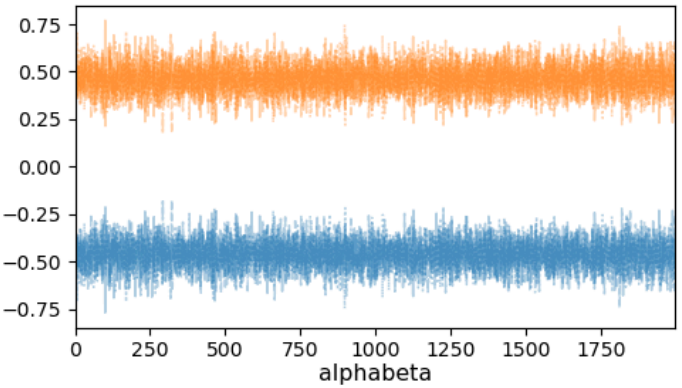
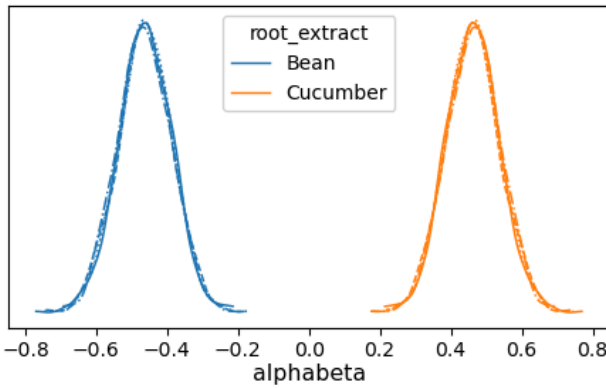
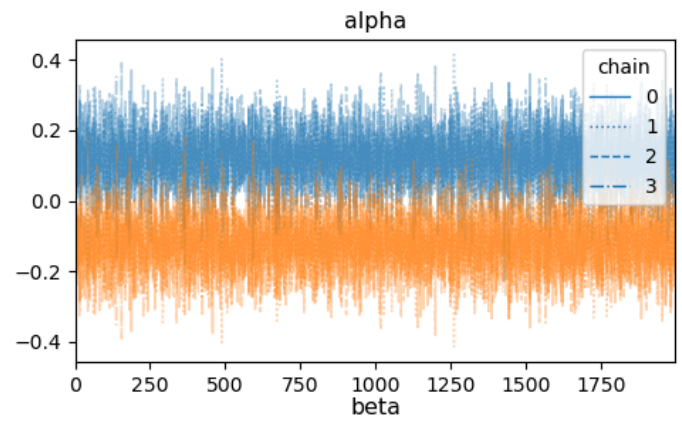
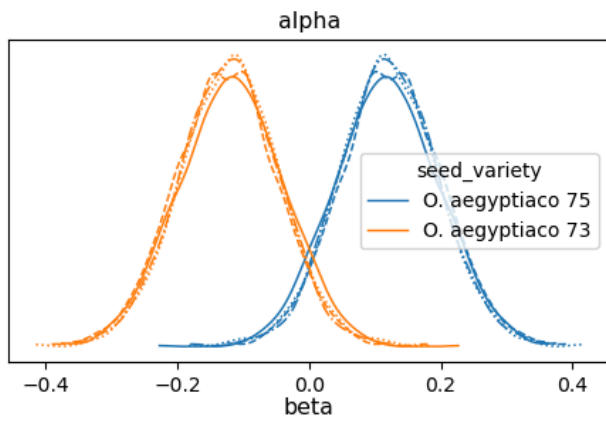


	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
oos_pred	61.75	8.738	45	78	0.11	0.069
mean x=2	61.72	4.042	53.746	69.547	0.073	0.042

Question 3

A linear model was created with an interaction term μ between the root extract and seed variety. A μ_0 standard normal distribution intercept was chosen and a Zero Sum Normal distribution $(0, 1)$ was assumed for the prior of the interaction term for simplicity in this factorial model.





	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alpha[O. aegyptiaco 75]	0.121	0.077	-0.023	0.278	0.001	0.001
alpha[O. aegyptiaco 73]	-0.121	0.077	-0.278	0.023	0.001	0.001
beta[Bean]	-0.462	0.076	-0.609	-0.314	0.001	0.001
beta[Cucumber]	0.462	0.076	0.314	0.609	0.001	0.001
alphabeta[O. aegyptiaco 75, Bean]	-0.196	0.078	-0.359	-0.057	0.001	0.001
alphabeta[O. aegyptiaco 75, Cucumber]	0.196	0.078	0.057	0.359	0.001	0.001

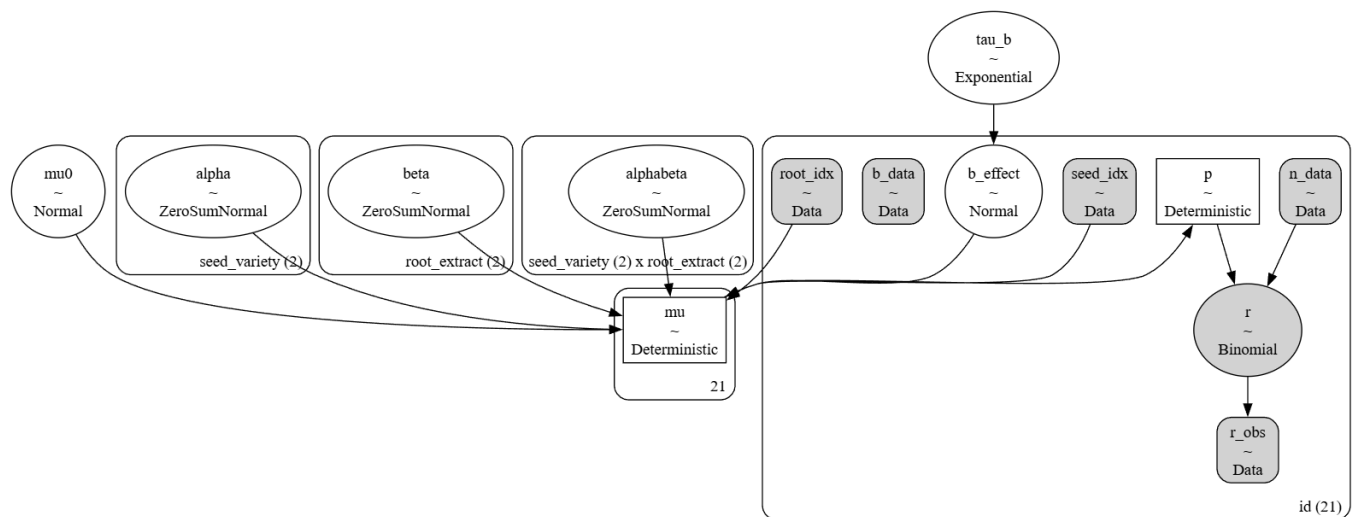
	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alphabeta[O. aegyptiaco 73, Bean]	0.196	0.078	0.057	0.359	0.001	0.001
alphabeta[O. aegyptiaco 73, Cucumber]	-0.196	0.078	-0.359	-0.057	0.001	0.001

$$R^2 = 0.90 \text{ \& } R^2 \text{ std} = 0.026$$

The model results show cucumber is associated with a higher germination rate probability than bean, although the particular combination of O. aegyptiaco 73 with Bean allows it to perform better than Bean on it's own.

Adding plate effect

A plate effect ("b_effect") was added that is normally distributed with variance, $\tau_b \sim \text{Exp}(5)$ Assuming an exponential distribution as value is positive and likely to be a smaller number. A λ value was chosen after running a few iterations and inspecting the forest plot, minimising the confidence intervals.



	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alpha[O. aegyptiaco 75]	0.218	0.174	-0.112	0.576	0.004	0.002
alpha[O. aegyptiaco 73]	-0.218	0.174	-0.576	0.112	0.004	0.002
beta[Bean]	-0.466	0.173	-0.817	-0.135	0.003	0.003
beta[Cucumber]	0.466	0.173	0.135	0.817	0.003	0.003

	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alphabet[a[O. aegyptiaco 75, Bean]	-0.198	0.17	-0.524	0.152	0.003	0.003
alphabet[a[O. aegyptiaco 75, Cucumber]	0.198	0.17	-0.152	0.524	0.003	0.003
alphabet[a[O. aegyptiaco 73, Bean]	0.198	0.17	-0.152	0.524	0.003	0.003
alphabet[a[O. aegyptiaco 73, Cucumber]	-0.198	0.17	-0.524	0.152	0.003	0.003

95.0% HDI

alpha[O. aegyptiaco 75]

[O. aegyptiaco 73]

beta[Bean]

[Cucumber]

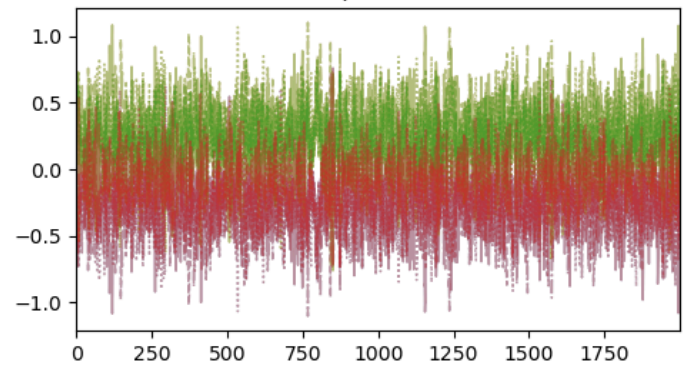
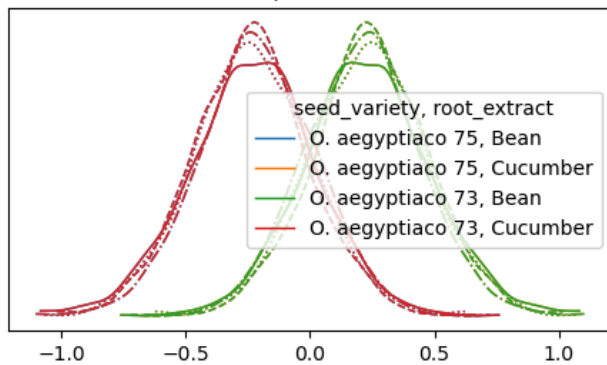
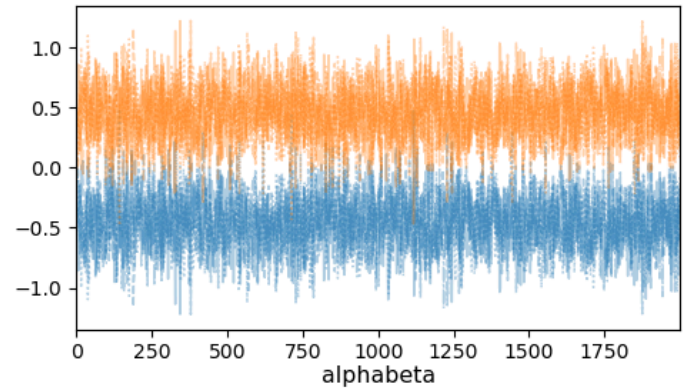
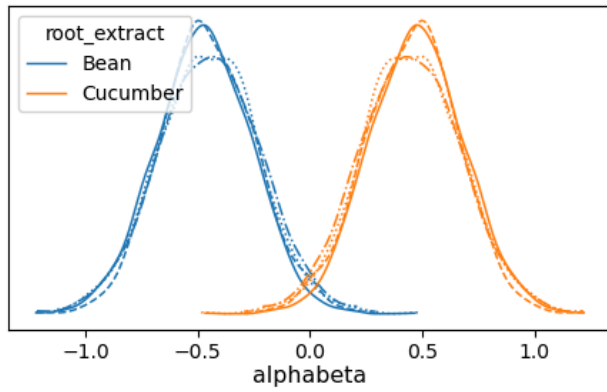
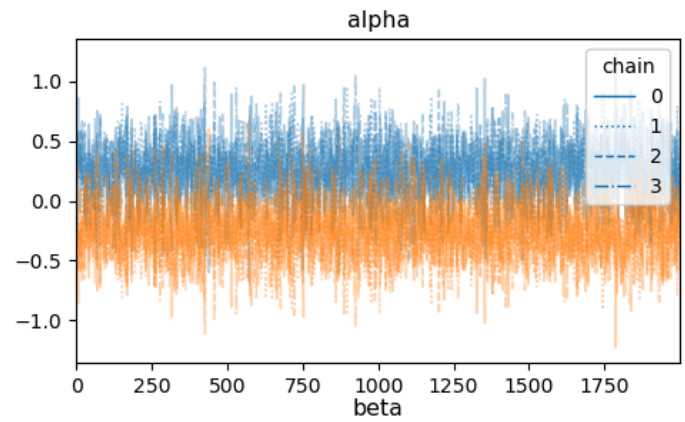
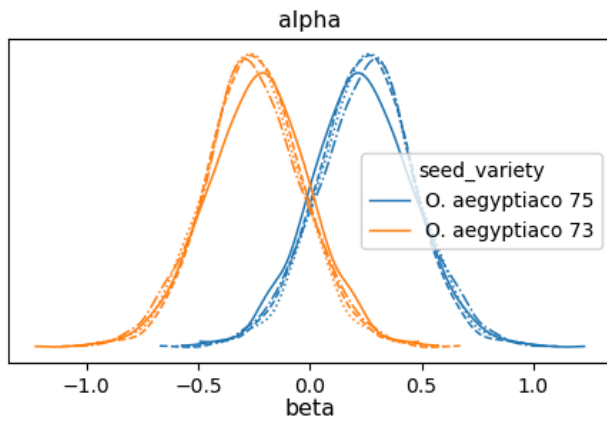
alphabetabeta[O. aegyptiaco 75, Bean]

[O. aegyptiaco 75, Cucumber]

[O. aegyptiaco 73, Bean]

[O. aegyptiaco 73, Cucumber]

-0.75 -0.50 -0.25 0.00 0.25 0.50 0.75



With plate effect: $R^2 = 0.94$. & $R^2_{std} = 0.021$

Adding this plate effect increased the R^2 value by 0.04 which is marginal and suggests the plate level effect doesn't significantly drive germination rates.

Interestingly, the standard deviation and HDI intervals have increased when adding the plate effect for all the parameters, meaning previous model (without plate effect) was perhaps overconfident in it's results and variability was not fully accounted for. Comparing the trace plots of the two models (with and without plate effects), this variability can also be clearly seen where the plate effect chains differ and vary more than without plate effect.

Note ESS values for all questions can be seen in the code and html file.